

SANJIT PRASAD

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Backend Engineer with 5+ years of experience building scalable distributed systems and industrial IoT platforms powering smart-city infrastructure, asset tracking, and real-time telemetry operations. Owned the architecture and production reliability of high-throughput event-driven pipelines processing 45M+ events/day across 10K+ connected devices, enabling energy optimization, incident automation, and operational visibility for municipal deployments. Strong expertise in Java, Spring Boot, Kafka, PostgreSQL, Redis, Kubernetes, and cloud-native backend systems.

TECHNICAL SKILLS

- **Languages/Frameworks:** Java, Spring Boot, JavaScript, Node.js, SQL
- **Distributed Systems & Backend:** Apache Kafka, MQTT, Event-Driven Architecture, Microservices, REST API Design, System Design, High Availability, Fault Tolerance, Scalability, Performance Optimization
- **Databases & Caching:** PostgreSQL, DynamoDB, Redis (Caching, Pub/Sub), Message Queues
- **Cloud & Infrastructure:** AWS (EC2, S3, SQS), Docker, Kubernetes, Cloud Infrastructure
- **DevOps & CI/CD:** Jenkins, Git, CI/CD Pipelines, Prometheus, Grafana, Production Monitoring
- **Domain:** IoT Platforms, Telemetry Systems, Asset Tracking, Smart Infrastructure

WORK EXPERIENCE:

Truminds Software Systems | *Software Development Engineer II*

October 2022 – Present

Bangalore, India

• Verso Altima (Croatian Government Smart Lighting Platform)

- * Architected and owned the fault-tolerant **Kafka + MQTT data pipeline** powering a municipal smart lighting platform built for Verso Altima, enabling real-time ingestion and processing of 45M+ telemetry events/day across 10,000+ smart poles and sustaining 99.9% uptime in production.
- * Designed and implemented a distributed event-driven pipeline on top of the in-house TruConnect platform to provide city-wide visibility into lighting infrastructure, allowing operational teams to monitor device health, automate incident generation, and improve maintenance response across multiple city deployments.
- * Solved data consistency issues between microservices and Kafka by implementing the **Transactional Outbox Pattern**, ensuring reliable event propagation for downstream analytics, alarms, and reporting systems critical for municipal decision-making.
- * Developed a configurable incident and reporting framework that enabled customers to detect device failures, telemetry anomalies, and inactivity issues faster, helping optimize field maintenance workflows and reduce infrastructure downtime.
- * Collaborated directly with business stakeholders and customer teams to understand energy optimization requirements and translate them into telemetry-driven reports, enabling better power-consumption analysis and measurable cost savings across smart-city operations.
- * Contributed to onboarding new smart lighting deployments across additional city infrastructures by improving platform scalability, integration flexibility, and device onboarding workflows.
- * Participated in production on-call rotations to troubleshoot live incidents, resolve telemetry ingestion failures, and maintain uninterrupted operational visibility for city-wide lighting infrastructure.

• TruConnect (Multi-tenant Industrial IoT Platform)

- * Owned backend architecture for TruConnect, a reusable Industrial IoT platform supporting smart lighting and smart logistics use cases, enabling customers to integrate and manage heterogeneous hardware providers through a unified operational dashboard.
- * Designed a provider abstraction layer for integrating **Qualcomm Aware** and Juganu APIs, solving inconsistent telemetry schemas, duplicate integration effort, and fragmented operational visibility across IoT vendors.
- * Optimized PostgreSQL query performance for telemetry-heavy dashboards using **indexing** and query-plan analysis, reducing **p95 latency** from multiple second to sub-second, improving dashboard responsiveness for internal operations and customers.
- * Led a PoC (Proof of Concept) integration for Juganu smart-lighting APIs into TruConnect, demonstrating seamless third-party onboarding and validating platform extensibility for future smart-city features such as digital signage, crowd analytics, and smart surveillance.

- **Qualcomm Smart Transportation System | Client Project**

- * Modernized a legacy employee transportation platform for Qualcomm used by employees, drivers, and dispatch teams to manage shuttle bookings, boarding workflows, and daily office commute operations.
- * Led the migration of push notification infrastructure from **Google Cloud Messaging to Firebase Cloud Messaging** ahead of deprecation deadlines, ensuring uninterrupted communication between drivers and employees and preventing service disruption.
- * Developed and enhanced mobile-facing backend features supporting employee ride bookings, driver updates, and dispatcher coordination, improving operational efficiency and commuter experience.

Tata Consultancy Services | Associate System Engineer

February 2021 – October 2022

Mumbai, India

- **Ericsson Telecom Platform Reliability**

- * Worked on distributed **ETCD**-backed microservices supporting telecom infrastructure systems for Ericsson, contributing to monitoring, bug fixes, production deployments, and feature enhancements across multiple backend services.
- * Built reusable **Grafana** dashboards for engineering teams to improve observability across 15+ microservices, reducing debugging effort and enabling faster identification of production bottlenecks.
- * Implemented an automated etcd inconsistency detection and remediation tool that prevented service downtime caused by cluster-state corruption, improving deployment stability and reducing dependency on senior SRE escalations.

EDUCATIONAL QUALIFICATIONS

Lakshmi Narain College of Technology and Science

August 2016 - June 2020

Bachelor of Technology in Electronics and Communication Engineering (8.00 CGPA)

Bhopal, MP, India